

Top 20 Dependency Injection Interview Questions (Spring Boot)

1 What is Dependency Injection?

Answer:

Dependency Injection is a design pattern where an object's dependencies are provided by an external container instead of being created by the object itself.

2 Why do we need Dependency Injection?

Answer:

To achieve:

- Loose coupling
 - Better testability
 - Easier maintenance
 - Cleaner code
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3 How does Spring implement Dependency Injection?

Answer:

Spring uses the **IoC (Inversion of Control) container** to create, manage, and inject objects (beans).

4 What is a Spring Bean?

Answer:

A bean is an object that is created, managed, and destroyed by the Spring container.

5 What are the types of Dependency Injection in Spring?

Answer:

1. Constructor Injection
 2. Setter Injection
 3. Field Injection (not recommended)
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6 Which Dependency Injection is recommended and why?

Answer:

✓ **Constructor Injection**

Because it:

- Ensures immutability
 - Makes dependencies mandatory
 - Is easier to test
 - Avoids NullPointerException
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7 What is @Autowired?

Answer:

@Autowired tells Spring to automatically inject a matching bean into a class.

8 Is @Autowired mandatory in Spring Boot?

Answer:

No.

Since Spring 4.3+, if a class has **only one constructor**, @Autowired is optional.

9 What is @Component?

Answer:

@Component marks a class as a Spring-managed bean.

10 Difference between @Component, @Service, and @Repository?

Answer:

Annotation	Purpose
@Component	Generic bean
@Service	Business logic layer
@Repository	DAO layer + exception translation

11 What is @Bean?

Answer:

@Bean is used to define a bean inside a **configuration class**, usually for third-party classes.

12 Difference between @Component and @Bean?

Answer:

@Component	@Bean
Used on class	Used on method
Auto scanned	Manually declared
Less control	Full control

13 What is @Configuration?

Answer:

Marks a class that contains @Bean methods and enables Spring to manage them properly.

14 What happens if multiple beans of same type exist?

Answer:

Spring throws:

`NoUniqueBeanDefinitionException`

15 How do you resolve multiple bean conflict?

Answer:

- `@Primary`
 - `@Qualifier`
 - Bean name injection
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16 What is @Primary?

Answer:

Marks a bean as the default choice when multiple beans of the same type exist.

17 What is @Qualifier?

Answer:

Used to explicitly specify which bean should be injected.

Example:

`@Qualifier("paypalPayment")`

18 What is circular dependency?

Answer:

When two or more beans depend on each other directly or indirectly.

Example:

A → B → A

19) How does Spring handle circular dependency?

Answer:

- Works with **setter injection**
 - Fails with **constructor injection**
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20) What is Inversion of Control (IoC)?

Answer:

IoC means control of object creation is transferred from the programmer to the Spring container.